

Exploration and Exploitation in a Distributed Representation of Organizational Knowledge

Muhammad A. Farooq and Guido Fioretti
University of Bologna
Computer Science Department and Management Department

May 2026

Abstract

We revisit March’s 1991 exploration-exploitation model by means of a recursive neural network in order to build a distributed representation of organizational knowledge. We confirm certain results of the original model and suggest additional insights.

1 Introduction

In 1991, James March introduced the concepts of *exploration* and *exploitation* into management studies with an article based on a computer simulation where organizational knowledge developed through interactions with individual knowledge and environmental stimuli.¹ Specifically, exploration was assumed to be carried out by stubborn individuals who pursued their own search for novelty in fields that they firmly believed would prove to be fruitful, whereas exploitation of available knowledge would imply adapting to novelties introduced elsewhere. Thus, explorers and exploiters could be distinguished by their speed of adjustment to organizational knowledge, with persistent explorers being slower than adaptive exploiters. The bulk of March’s simulations concerned the interplay of individual and organizational knowledge, where the differential speed of explorative and imitative organization members generated non-obvious insights concerning the development of an organization’s knowledge and its ability to deal with an uncertain environment [26].

The concepts of exploration and exploitation have been immensely successful, though their interpretations have deviated substantially from the scope that James March had originally conceived [2]. Rather than focusing on organizational learning, substantial streams of research asked what environmental

¹March [26] actually employed the term “organizational code” for organizational knowledge.

conditions would either prompt exploration or exploitation [35], or possibly a sequence of exploration and exploitation [12], or arguing for a systemic balance between explorers and exploiters in organizational ecosystems [13]. The original orientation towards internal organizational dynamics can still be found in studies on *ambidexterity*, by definition the ability to carry out both exploration and exploitation at a time. In particular, most research on ambidexterity has focused on managerial practices and organizational design, with a substantial stream investigating the interplay of exploration and exploitation in terms of network structures [27] [25] [36]. This approach is possibly closest to the focus on optimal balance between exploration and exploitation that characterized March’s 1991 article [26], but the relation between individual exploration or exploitation on the one hand, and organizational knowledge on the other hand, has been lost. Network models focus on the relations between single organization members, but do not think of organizational knowledge as an asset that goes beyond the aggregation of individual knowledge [37] [17].

Our work is inspired by the connectionist understanding of brains as organs that are capable of generating phenomena that transcend their component neurons because those neurons allow information to be retained insofar it circulates in loops [16]. This sort of distributed memories, whose content does not reside at any specific place and yet exists insofar it circulates, can be found in as diverse objects as autocatalytic reaction networks, insect societies and the immune system [8] [28]. Following an insight originating from Hayek [38] [7], we maintain that the same principle can also be at work in human organizations and societies at large, with individuals taking the place of neurons [22] [15]. Specifically, we understand such information loops as organizational routines [29] [30], where in quite many cases the loop closes through other organizations (e.g., customers and suppliers).

We make use of this analogy *cum grano salis*. Specifically, we employ a computational model where information loops originate from the need to predict environmental signals, as it is indeed the case in March’s model [26]. While confirming the basic results concerning the importance of exploration, we are able to point to additional properties that are invisible to an aggregated representation of organizational knowledge. In particular, we TO BE FINISHED

The rest of this article is organized as follows. DDDDD D DYTDF YTF YFG UOGY IUG IPUH PIOUH OIHJ OÈI UJOPIJ PJ IOPJ JIOP POJ IOPJ

2 Organizational Routines as Information Loops

Connectionism has revolutionized cognitive sciences by focusing on the paths taken by neuron signals, rather than assuming that algorithms constitute the basis of intelligence [16] [5]. In this paradigm information loops have a special place, because they enable the formation and storage of concepts on the substrate of neuronal activity [14] [24] [11].

Several researchers have mentioned the possibility of applying the philosophy and the tools of connectionism to human organizations, with humans in place

of neurons and organizational routines as information loops [22] [15]. This conceptual mapping makes sense for several reasons.

Firstly, there already exists a stream of literature that identifies routines in terms of sequences of activities, eventually closing in a loop [29] [30]. Connectionism would add the insight that being closed in a loop is mandatory for a routine to sustain itself, with substantial portions of the loop eventually flowing through other organizations such as customers or suppliers.

Secondly, just as information loops in the brain seldom flow through one and the same path again and again [34] [10], so do organizational routines [31] [32]. The stability of organizational routines pertains to their ostensive character, whereas their similarity with connectionism concerns their performative level [9].

Thirdly, albeit the brain has hierarchical structures that exert some control over information flows, many bifurcations are taken depending on local conditions and limited, local information. In organization science, this resonates with nuanced understandings of organizations [3], as well as indirect ways of influencing subordinates behavior ranging from lean manufacturing [23] to online communities developing free software [6] or *Wikipedia* [1], to flat organizations with a minimal number of hierarchical levels [4]. In the limit, organizational routines may be developed by actors who are not even aware of the pattern of actions that they are generating [19] [20] [33]. To the extent that this limiting point is approached, the agency of individuals in organizations is comparable to that of neurons in a brain.

Artificial Neural Networks (ANNs) are the basic tool of connectionism. In particular, Recurrent Neural Networks (RNNs) entail feed-backs that enable information to loop within a network. In our conceptual mapping, such loops represent routines that are entirely contained within one organization.

ANNs reproduce the basic cognitive activity of classifying incoming information into mental categories. In particular, categories are conceived as building by adding examples around a prototype. To make a simple example, a child may be first exposed to a kitchen chair, adding instances such as “armchair” or “office chair” later on.

ANNs exploit the many-to-one property of addition (as e.g., 4 can result from $2 + 2$ as well as $3 + 1$, etc.). A *neuron* generates output y by weighing inputs $x_1, x_2, \dots, x_n$ by means of weights $w_1, w_2, \dots, w_n$:

$$y = \sum_i w_i x_i \quad (1)$$

A neural network entails several neurons. Figure 1 illustrates an ANN made of one single layer of neurons. This network operates a linear projection of input $\mathbf{x}$ into output $\mathbf{y}$:

$$\mathbf{y} = \mathbf{W} \mathbf{x} \quad (2)$$

where $\mathbf{W}$ is a matrix with as many rows as neurons and as many columns as weights per neuron.

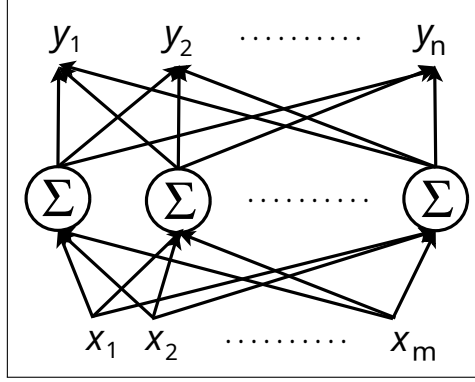


Figure 1: An ANN made of one single layer of neurons. Each neuron takes m inputs and generates one scalar output. With n neurons, the network operates $\mathbf{y} = \mathbf{W}\mathbf{x}$ where $\mathbf{x}$ is $m \times 1$, $\mathbf{y}$ is $n \times 1$ and $\mathbf{W}$ is $n \times m$.

Categories do not correspond to specific neurons. In general, one category rests on input information flowing along multiple paths through several neurons.

However, neurons typically specialize into firing a high output y when specific input patterns arrive. Specialization takes place by tuning weights w_i .

Quite often, the output of neurons is multiplied by the *sigmoid* function in order to constrain it within the $[0, 1]$ interval:

$$\sigma(y) = \frac{1}{1 - e^{-y}} \quad (3)$$

where y is meant to be the output of a neuron as in eq. 1.

In order to extract the sign of a signal, constraining it within the $[-1, 1]$ interval, the hyperbolic tangent function is eventually employed:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} \quad (4)$$

One common problem with RNNs is that information loops either explode or fade out with time. Long- Short-Time Memory cells (LSTMs) overcome this problem by ensuring that the amplitude of the long-term memory is kept roughly constant by balancing what old information is discarded with the new information that is added to the cell [18].

Figure 2 illustrates a simplified LSTM cell. The *Forgetting Gate* and the *Input Gate* are ANNs that are fed with the same input but employ different weights in order to compute what portion of the old information should be retained and what portion of the new incoming information should be combined with the previous one, respectively. The *Output Gate* is a third ANN that computes the output.

In a LSTM cell, the *hidden state* $\mathbf{h}$ expresses the short-term content of the memory, strongly influenced by the incoming signals. The *cell state* $\mathbf{c}_t$ expresses

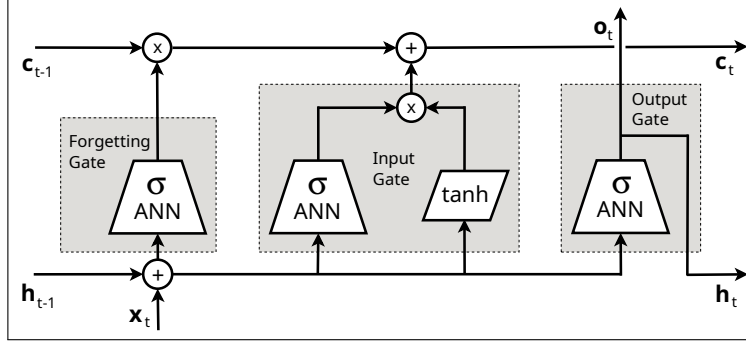


Figure 2: A simplified LSTM cell entails three ANNs, whose outputs are curbed by sigmoid functions. In the Input Gate, a hyperbolic tangent function is employed to determine the sign of the output of the ANN. At step t , the exogenous input $\mathbf{x}_t$ meets the cell state is $\mathbf{c}_{t-1}$ and the hidden state is $\mathbf{h}_{t-1}$ to generate output $\mathbf{o}_t$. In this simplified scheme, $\mathbf{o}_t \equiv \mathbf{h}_t$.

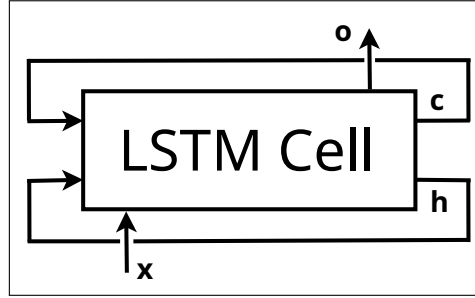


Figure 3: The feed-back loops of one single LSTM cell. Hidden state $\mathbf{h}$ and cell state $\mathbf{c}$ are fed back into the call.

the long-term content of the memory, that is impaired from either exploding or fading out. The output $\mathbf{o}$ is designed to predict the input. In the simplified scheme of Figure 2, it is equivalent to the hidden state.

One single LSTM is a RNN because the hidden state $\mathbf{h}_t$ and the cell state $\mathbf{c}_t$ that it generates at time t are fed back into the cell at time $t + 1$. Figure 3 illustrates the loops enabled by one single cell. In general, several LSTM cells are combined into a larger RNN.

LSTM-based RNNs are designed to predict the values that its input will receive in the next step. This is also the assumption of March 1991 model [26], reflecting the idea that predicting their environment is essential for organizations to survive.

Neuron weights are updated by means of a gradient descent algorithm that maximizes LSTM predictive ability. LSTM operation goes through the following steps:

1. **Forward Pass** computes the output;
2. **Loss Calculation** measures the error;
3. **Backward Pass** calculates the gradients for all weights;
4. **Optimizer Step** updates weights based on those gradients.

3 Exploration and Exploitation Embedded in Distributed Knowledge

We employ one single LSTM cell to represent one single organization. Just like in March’s model, environmental stimuli consist of sequences of vectors of length $D = 30$. Each sequence entails 200 vectors in as many time steps.² The hidden state is also a vector of length $D = 30$.

² LR

We let our LSTM go through this cycle feeding it with the following inputs:

1. Random integers drawn from a uniform distribution in $\{-1, 1\}$.
2. Random real numbers drawn from a uniform distribution in $[-1, 1]$.
3. A slow and predictable input represented by a sinusoid of frequency 1, amplitude 1, phase increasing by $2\pi/D$ for each component of the input vectors.
4. A fast but predictable input, represented by a sinusoid of frequency 10, amplitude 1, phase increasing by $2\pi/D$ for each component of the input vectors.
5. A slow and predictable input as in case 3 above, followed by a random input as in case 2 above.
6. A fast and predictable input as in case 4 above, followed by a random input as in case 2 above.
7. A slow and predictable input as in case 3 above, followed by a random input as in case 2 above, followed again by a slow and predictable input as in case 3 above.
8. A slow but predictable input as in case 4 above, followed by a random input as in case 2 above, followed again by a fast but predictable input as in case 4 above.

For each of these configurations, we observe:

²In March 1991 model [26], the number of vectors in the sequence was not specified.

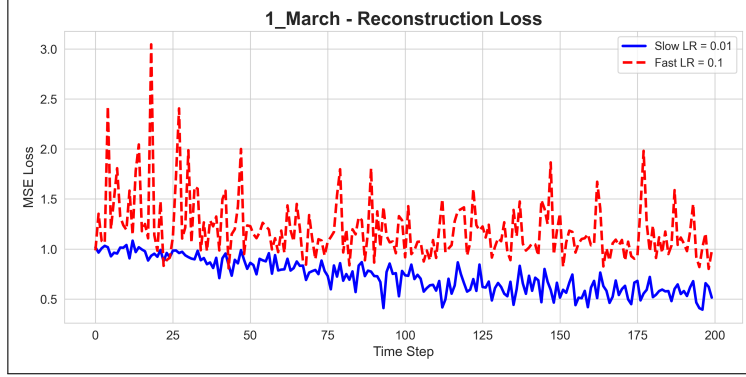


Figure 4: Reconstruction Loss with a random input drawn from $\{-1, 1\}$ with a uniform distribution. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

Reconstruction Loss The Mean Squared Error (MSE) between predicted values and actual values. The lower the MSE, the better the organization fares.

Hidden State Norm The magnitude of $\mathbf{h}_t$, the hidden state of the LSTM cell. The hidden state represents the organization’s short-term memory. Since it is largely obtained from neurons’ reactions to current input, it can be interpreted as an average of individual knowledge.

Cell State Norm The magnitude of $\mathbf{c}_t$, the cell state. This represents the long-term memory of the cell. It can be interpreted as organizational memory, organizational knowledge, or culture.

3.1 Random Integers drawn from a Uniform Distribution

This input is a random sequence drawn from $\{-1, 1\}$. This may have been the input of March 1991 model [26] if he mentioned “probability” in terms of time but, in light of his later discussion of “turbulent” environments, it rather seems that he simply employed a sequence of identical input vectors, which we did not even attempt to feed into the network.

Figure 4 illustrates the reconstruction loss for slow learners (blue line) and fast learners (red dashed line), respectively. The lower the reconstruction loss, the better the network predicts the incoming signal.

One property of March 1991 model was that individual and collective knowledge tend to converge over time. Figure 4 shows that LSTM predictions become increasingly accurate. At this stage, we speculate that this happens because individual and organizational knowledge converge towards an improved understanding of reality.

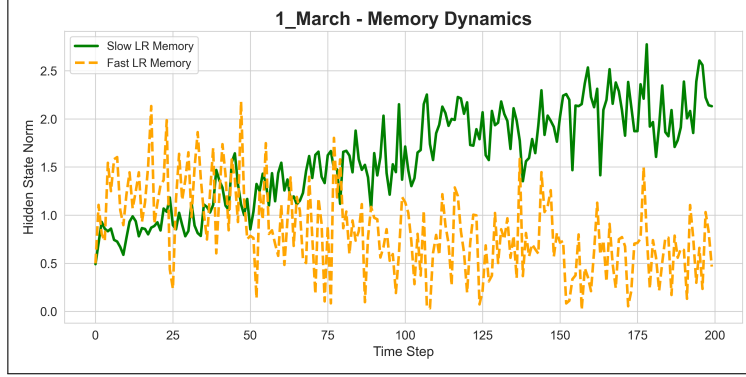


Figure 5: Hidden State Norm with a random input drawn from $\{-1, 1\}$ with a uniform distribution. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

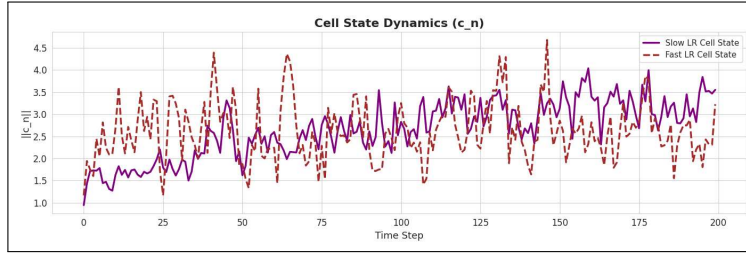


Figure 6: cELL State Norm with a random input drawn from $\{-1, 1\}$ with a uniform distribution. Purple continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

Figure 5 illustrates the hidden state norm (amplitude) for slow learners (green continuous line) and fast learners (yellow dashed line), respectively. The hidden state represents the short-term memory of the LSTM, largely influenced by individual neurons.

Figure 6 illustrates the cell state norm (amplitude) for slow learners (purple continuous line) and fast learners (red dashed line), respectively. The cell state represents the long-term memory of the LSTM, corresponding to organizational knowledge.

March 1991 claimed that equilibrium knowledge is highest when either (i) individual learning is slow whereas organizational learning is fast, or (ii) individual learning is fast whereas organizational learning is slow. Furthermore, the arrangement (i) is more efficient than (ii).

Insofar slow individual learning signals exploration whereas fast individual learning signals exploitation, property (i) means that the most efficient organizations value the findings of explorative individuals, whereas (ii) means that with individual organization members mostly devoted to exploit available pos-

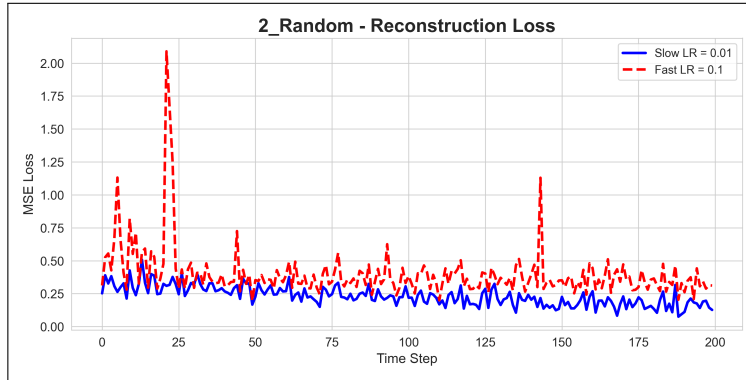


Figure 7: Reconstruction Loss with a random input drawn from $[-1, 1] \in \mathbb{R}$ with a uniform distribution. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

sibilities, the most efficient organizations are those that are cautious with incorporating individual knowledge. Thus, March claimed that the most efficient organizations are those with many explorative individuals, and an organizational culture that quickly values them. Conversely, insofar it concerns organizations that are mostly populated by exploitative individuals, the highest efficiency is achieved when the organization is slow in following their individual members.

Let us compare Figures 4, 5 and 6. All figures have been obtained by minimizing MSE reconstruction loss given the constraint of either slow (explorative) or fast (exploitative) learners.

The slow, explorative learners (continuous lines) achieve higher efficiency, much larger individual knowledge, slightly larger and possibly less variable organizational knowledge. On the whole, March’s insights surface again.

3.2 Random real numbers drawn from a uniform distribution

With random real numbers drawn from the $[-1, 1]$ interval, no qualitative difference is apparent in comparison to the input of integers considered in § 3.1. Figures 7 and 8 are qualitatively equivalent to 4 and 5, respectively.

3.3 A slow sinusoid

The organization is supposed to received an easily predictable input signal. If March initially fed his model with a constant vector, then this is the input that most closely approximates his initial input.

Figure 9 illustrates the reconstruction loss with a slow, predictable, sinusoidal input. With slow learners, it stays nearly always at zero. Only fast learners experience occasional bursts of higher reconstruction loss, signaling

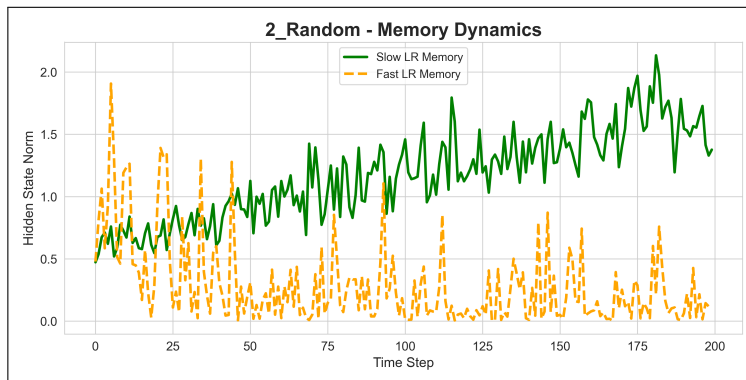


Figure 8: Hidden State Norm with a random input drawn from $[-1, 1] \in \mathfrak{R}$ with a uniform distribution. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

once again the inefficiency of exploiters.

This dynamics reflects very much an insight that was already mentioned by March [26], namely the observation that organizational learning is less efficient in turbulent environments. In comparison to this sinusoid, the random environment of § 3.1 and 3.2 was more turbulent and, indeed, the reconstruction loss was definitely higher.

Figure 10 illustrates the hidden state norm (short-term organizational distributed memory) with a slow sinusoidal input. We still find that slow-learners contribute more to organizational knowledge, but the difference is less clear-cut than in the case of random input. With predictability, fast-learning exploiters do not fare so badly as in purely random environment.

However, comparing slow and fast learners in Figures 9 and 10 suggests that although fast learners may be close to slow learners insofar it concerns their contribution to organizational knowledge, they may occasionally fare very poorly in terms of reconstruction loss.

3.4 A fast sinusoid

By exposing our LSTM with a fast sinusoid representing a not-so-easily predictable environment, we are considering a case that is different from those envisioned by March [26]. Let us feed the LSTM a sinusoid with frequency ten times as high as that of § 3.3.

Figure 11 illustrates the reconstruction loss with a fast sinusoidal input. While reconstruction loss stays nearly at zero for slow learners, fast learners focus on short-term oscillations. Thus, they are much less efficient.

According to Figure 12, average short-term organizational memory may not be so different after all, but fast-learners generate exceedingly high oscillations. In organizations pursuing exploitation only, organizational knowledge oscillates

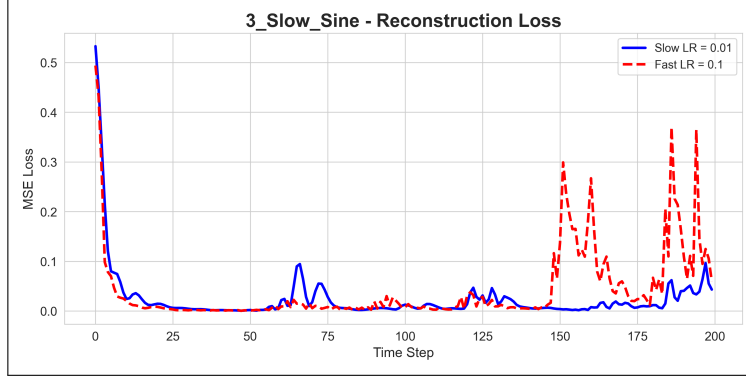


Figure 9: Reconstruction Loss with a slow sinusoidal input, frequency equal to 1. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

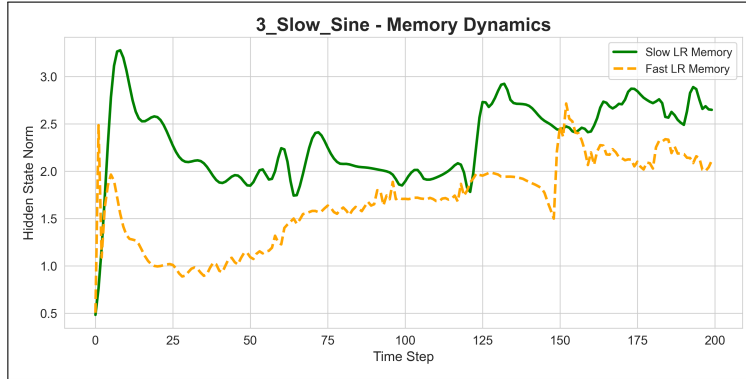


Figure 10: Hidden State Norm with a slow sinusoidal input, frequency equal to 1. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

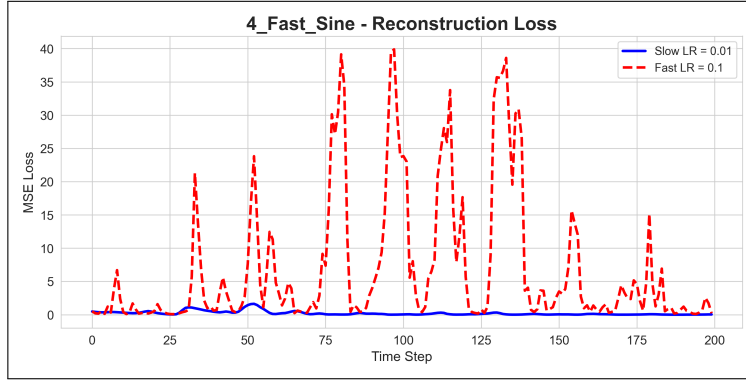


Figure 11: Reconstruction Loss with a fast sinusoidal input, frequency equal to 10. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

and their ability to predict their environment oscillates accordingly.

3.5 A slow sinusoid followed by a random input

In this section we feed our LSTM first with a slow sinusoid as in § 3.3, then a random input as in § 3.2. The idea is that of seeing whether knowledge of an easily detectable pattern worsens or improve reactions to an unpredictable environment.

Figure 13 illustrates the reconstruction loss. The interesting feature is that fast-learners apparently have a harder time than slow-learners to adjust to the random input.

However, Figure 14 shows that this initial inefficiency does not correspond to any specific feature of organizational short-term knowledge. It is not a matter of amplitude, rather the phase of oscillations.

3.6 A fast sinusoid followed by a random input

Figures 15 and 16 are the counterparts of 13 and 13, respectively. Contrary to § 3.5, the fast sinusoid does not hinder the transition of fast-learners to the random input.

3.7 A slow sinusoid followed by a random input followed by a slow sinusoid again

With this input sequence, we want to investigate whether having learned a slow sinusoidal pattern, the LSTM responds better when this pattern appears again after a chaotic interruption. Apparently, this may be the case of the distributed knowledge of organizations populated by slow-learners, but not of

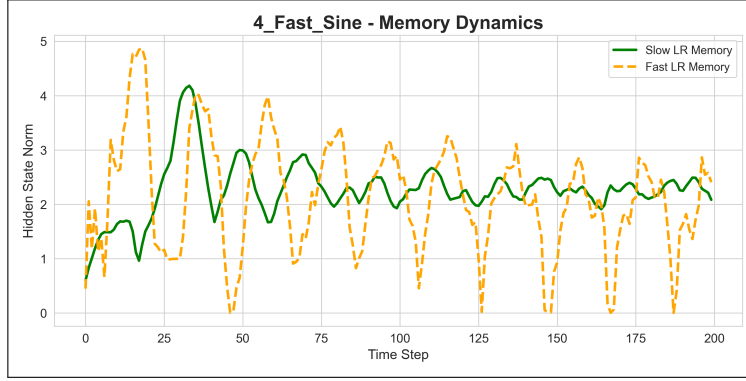


Figure 12: Hidden State Norm with a fast sinusoidal input, frequency equal to 10. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

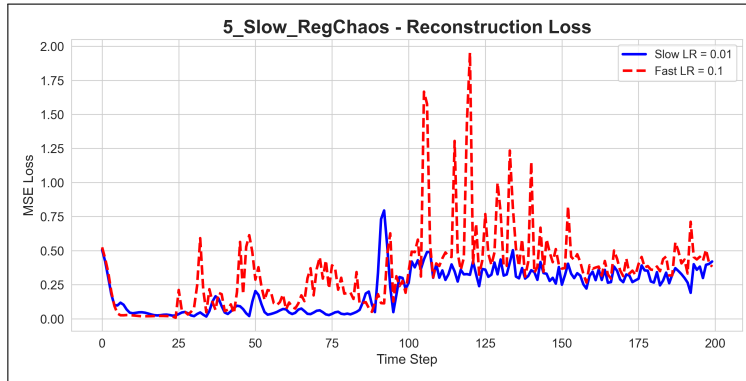


Figure 13: Reconstruction Loss with a slow sinusoidal input, frequency equal to 1, followed by a random input. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

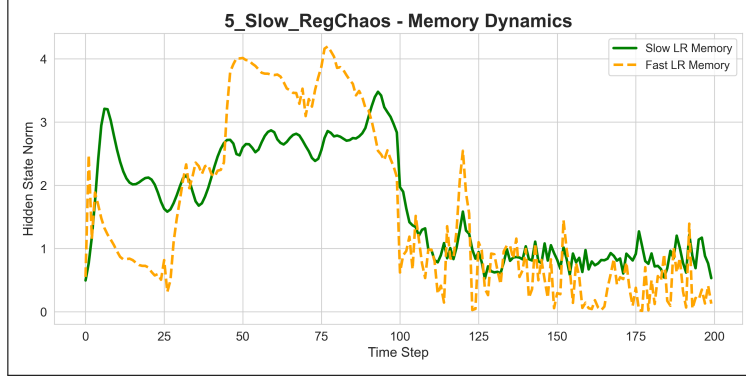


Figure 14: Hidden State Norm with a slow sinusoidal input, frequency equal to 1, followed by a random input. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

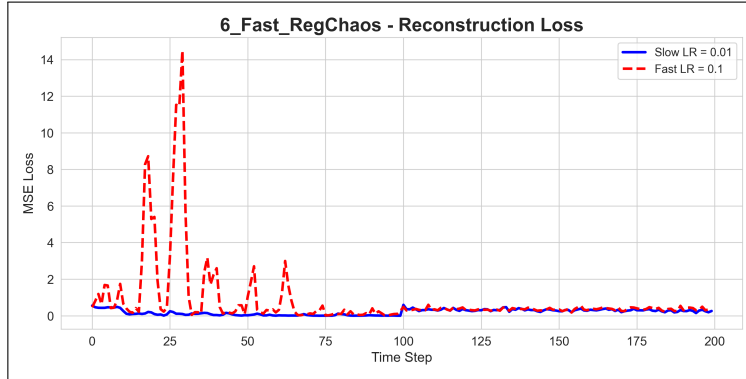


Figure 15: Reconstruction Loss with a fast sinusoidal input, frequency equal to 10, followed by a random input. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

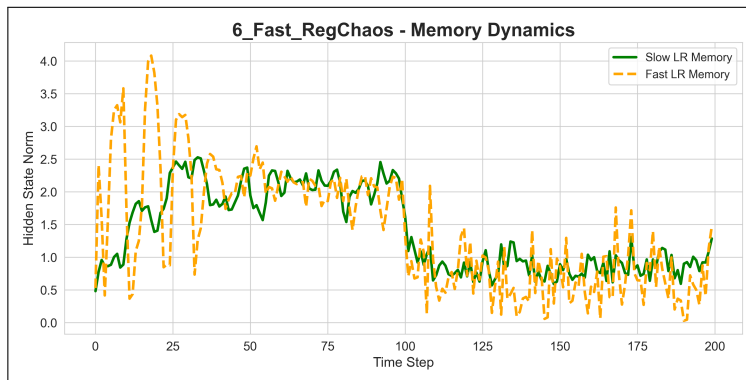


Figure 16: Hidden State Norm with a fast sinusoidal input, frequency equal to 10, followed by a random input. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

organizations populated by fast-learners. Notably, this difference has no impact on reconstruction loss.

3.8 A fast sinusoid followed by a random input followed by a fast sinusoid again

Same input pattern as in § 3.7, but with a fast sinusoid. Contrary to the previous case, no signs that the LSTM learned patterns that it recognized after some interruption.

4 Next Steps

This paper is a very preliminary exploration of what could be done with LSTMs. In this article, an analogy has been drawn between a human organization and one single LSTM cell, with neurons in the cell playing the role of humans. A different analogy would be possibly more appropriate, one where one LSTM cell represents one single individual, and a network of LSTM cells represents the organization. In this case, the neurons within each LSTM cell would represent brain neurons.

This alternative analogy would have at least two advantages. First, it is easier to make the components of an LSTM cell correspond to different functions in the brain, rather than different offices or roles within an organization. Secondly, the structure of connections between LSTM cells would correspond to communication structures of individuals within an organization.

However, *PyTorch* and other libraries set clear limits on what can be currently done. It is possible to arrange several LSTM cells in a stack, allowing bi-directional communication between neighboring cells as illustrated in Figure 21. This would correspond to individuals in a line, talking to one another.

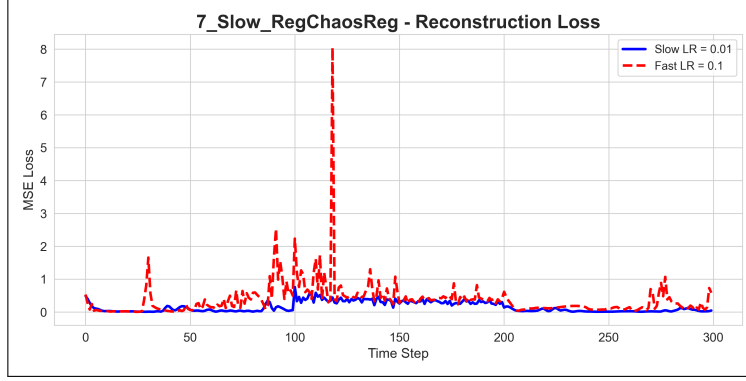


Figure 17: Reconstruction Loss with a slow sinusoidal input, frequency equal to 1, followed by a random input, followed again by a slow sinusoidal input. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

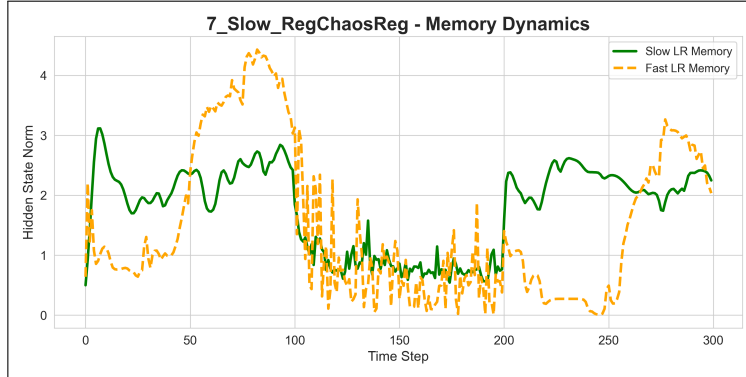


Figure 18: Hidden State Norm with a slow sinusoidal input, frequency equal to 1, followed by a random input, followed again by a slow sinusoidal input. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

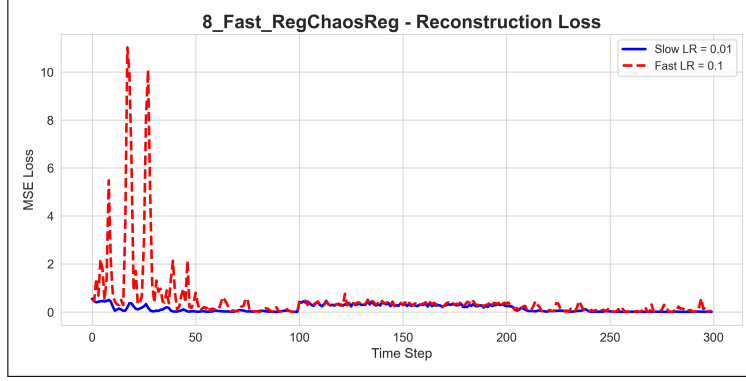


Figure 19: Reconstruction Loss with a fast sinusoidal input, frequency equal to 1, followed by a random input, followed again by a fast sinusoidal input. Blue continuous line: Slow Learning Rate $LR = 0.01$. Red dashed line: Fast Learning Rate $LR = 0.1$.

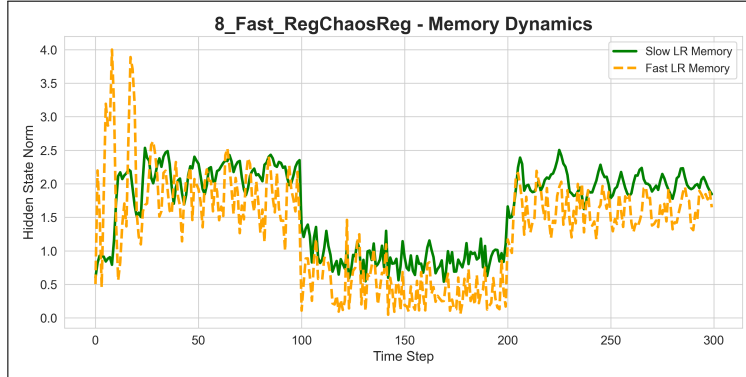


Figure 20: Hidden State Norm with a fast sinusoidal input, frequency equal to 1, followed by a random input, followed again by a fast sinusoidal input. Green continuous line: Slow Learning Rate $LR = 0.01$. Yellow dashed line: Fast Learning Rate $LR = 0.1$.

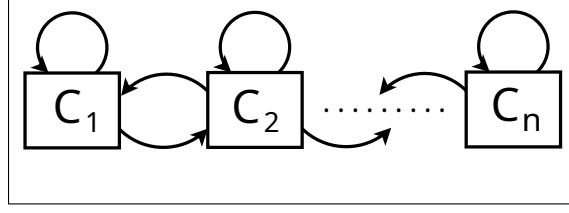


Figure 21: A stack of LSTM cells. Loops exist both between cells as from each cell onto itself.

However, it would not be possible to explore other arrangements, e.g., hierarchies of LSTM cells.

One other, very important extension would be allowed by LSTMs that entail unsupervised ANNs, capable of updating their own weights by means of some form of Hebbian rule. This would enable the simulator to identify and recognize input patterns, which corresponds to organizations resorting to specific routines when they face specific problems.

Legal Disclaimers

This article does not entail copyright materials. Neither human nor animal subjects were involved.

Authors are listed in alphabetical order. GF contributed conceptualization and writing. MAF wrote the simulation code and carried out the experiments.

A The *PyTorch* LSTM

We used the LSTM made available by the *PyTorch* library. The *PyTorch* LSTM entails eight ANNs. Four ANNs receive input $\mathbf{x}_t$, four ANNs receive input $\mathbf{h}_t$. Omitting biases ³ for simplicity, this LSTM is described by the following equations:

$$\mathbf{i}_t = \sigma(\mathbf{W}^{ii} \mathbf{x}_t + \mathbf{W}^{hi} \mathbf{h}_{t-1}) \quad (5)$$

$$\mathbf{f}_t = \sigma(\mathbf{W}^{if} \mathbf{x}_t + \mathbf{W}^{hf} \mathbf{h}_{t-1}) \quad (6)$$

$$\mathbf{g}_t = \tanh(\mathbf{W}^{ig} \mathbf{x}_t + \mathbf{W}^{hg} \mathbf{h}_{t-1}) \quad (7)$$

$$\mathbf{o}_t = \sigma(\mathbf{W}^{io} \mathbf{x}_t + \mathbf{W}^{ho} \mathbf{h}_{t-1}) \quad (8)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \mathbf{g}_t \quad (9)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (10)$$

³In LSTMs, *biases* are trainable parameters that shift the output of activation functions σ and $\tanh$ horizontally, allowing information to flow even when the input is proximate to zero.

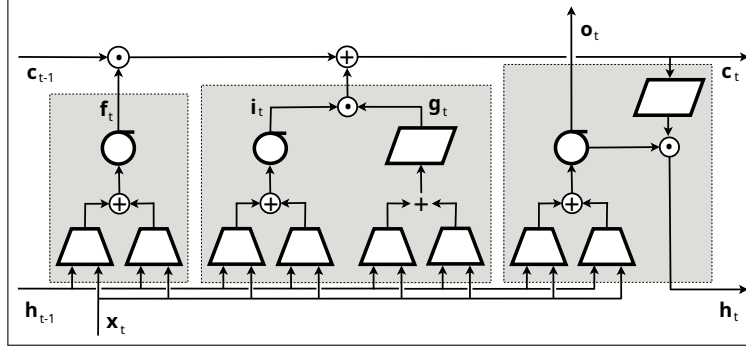


Figure 22: The architecture of the *PyTorch* LSTM. The three shaded blocks are the Forgetting Gate, the Input Gate and the Output Gate, respectively. Trapezoids represent ANNs, sigmas represent sigmoids, rhomboids represent hyperbolic tangents.

where $\mathbf{h}_t$ is the hidden state at time t , $\mathbf{c}_t$ is the cell state at time t , $\mathbf{x}_t$ is the input at time t , $\mathbf{h}_{t-1}$ is the hidden state of the layer at time $t - 1$ or the initial hidden state at time 0, and $\mathbf{i}_t$, $\mathbf{f}_t$, $\mathbf{g}_t$, $\mathbf{o}_t$ are the signals generated by the input, forget, cell, and output gates, respectively. Matrices $\mathbf{W}^{pq}$ entail the coefficients of ANNs that receive input p and operate in gate q , with $p \in \{\mathbf{i}, \mathbf{h}\}$ and $q \in \{\mathbf{i}, \mathbf{f}, \mathbf{g}, \mathbf{o}\}$. These ANNs generate outputs that are curbed by either the sigmoid function σ , or the hyperbolic tangent function $\tanh$. The symbol $\odot$ denotes the Hadamard product.

Note that this LSTM is more complex than the example illustrated in Figure 2. Figure 22 illustrates its architecture.

References

- [1] Aleksi Aaltonen and Gian Francesco Lanzara. Building governance capability in online social production: Insights from wikipedia. *Organization Studies*, 36(12):1649–1673, 2015.
- [2] Ranga Almahendra and Björn Ambos. Exploration and exploitation: A 20-year review of evolution and reconceptualization. *International Journal of Innovation Management*, 19(01):1550008, 2015.
- [3] Chester I. Barnard. *The Functions of the Executive*. Harvard University Press, Boston, 1938.
- [4] Giacomo Cabri and Guido Fioretti. Flexibility out of standardization. *International Journal of Organization Theory & Behavior*, 25(1–2):22–38, 2022.
- [5] Andy Clark. *Associative Engines: Connectionism, Concepts, and Representational Change*. The MIT Press, Cambridge, 1993.

- [6] Paul B. de Laat. Governance of open source software: State of the art. *Journal of Management & Governance*, 11(2):165–177, 2007.
- [7] Massimo Egidi and Salvatore Rizzello. Cognitive economics: Foundations and historical evolution. In Massimo Egidi and Salvatore Rizzello, editors, *Cognitive Economics*, chapter I, pages 1–22. Edward Elgar, Cheltenham, 2004.
- [8] J. Doyne Farmer. A rosetta stone for connectionism. *Physica D*, 42(1-3):153–187, 1990.
- [9] Martha S. Feldman and Brian T. Pentland. Reconceptualizing organizational routines as a source of flexibility and change. *Administrative Science Quarterly*, 48(1):94–118, 2003.
- [10] Andrew A. Fingelkurts and Alexander A. Fingelkurts. Information flow in the brain: Ordered sequences of metastable states. *Information*, 8(1):22, 2017.
- [11] Joaquín M. Fuster. Cortex and memory: Emergence of a new paradigm. *Journal of Cognitive Neuroscience*, 21(11):2047–2072, 2009.
- [12] Victor Gilsing and Bart Nooteboom. Exploration and exploitation in innovation systems: The case of pharmaceutical biotechnology. *Research Policy*, 35(1):1–23, 2006.
- [13] Anil K. Gupta, Ken G. Smith, and Christina E. Shalley. The interplay between exploration and exploitation. *Academy of Management Journal*, 49(4):693–706, 2006.
- [14] Stevan Harnad. On the nature of explanation: A pdp approach. *Physica D*, 42(1-3):335–346, 1990.
- [15] Jonas Hauke, Iris Lorscheid, and Matthias Meyer. Individuals and their interactions in demand planning processes: An agent-based, computational testbed. *International Journal of Production Research*, 56(13):4644–4658, 2018.
- [16] Donald O. Hebb. *The Organization of Behavior. A Neuropsychological Theory*. John Wiley & Sons, New York, 1949.
- [17] Achim Hecker. Knowledge beyond the individual? making sense of a notion of collective knowledge in organization theory. *Organization Studies*, 33(3):423–445, 2012.
- [18] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997.
- [19] Edwin Hutchins. Organizing work by adaptation. *Organization Science*, 2(1):14–39, 1991. Reprinted in [21].

- [20] Edwin Hutchins. *Cognition in the Wild*. The MIT Press, Cambridge, 1995.
- [21] Edwin Hutchins. Organizing work by adaptation. In Michael D. Cohen and Lee S. Sproull, editors, *Organizational Learning*, chapter II, pages 20–57. Sage Publications, Thousand Oaks, 1996.
- [22] Norman L. Johnson, Steen Rasmussen, Cliff Joslyn, Luis Rocha, Steven Smith, and Marianna Kantor. Symbiotic intelligence: Self-organizing knowledge on distributed networks driven by human interaction. In Christoph Adami, Richard K. Belew, Hiroaki Kitano, and Charles E. Taylor, editors, *Artificial Life VI: Proceedings of the Sixth International Conference on Artificial Life*, pages 403–407, Cambridge (MA), 1998. The MIT Press.
- [23] Robert Bruce Johnston. Making manufacturing practices tacit: A case-study of computer-aided production management and lean production. *The Journal of the Operational Research Society*, 46(10):1174–1183, 1995.
- [24] Peter König and Norbert Krüger. Symbols as self-emergent entities in an optimization process of feature extraction and prediction. *Biological Cybernetics*, 94(4):325–334, 2006.
- [25] David Lazer and Allan Friedman. The network structure of exploration and exploitation. *Administrative Science Quarterly*, 52(4):667–694, 2007.
- [26] James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1982.
- [27] Kent D. Miller, Meng Zhao, and Roger J. Calantone. The network structure of exploration and exploitation. *Academy of Management Journal*, 49(4):709–722, 2006.
- [28] Jordi Pi nero and Ricard Solé. Statistical physics of liquid brains. *Philosophical Transactions B*, 374(20180376), 2019.
- [29] Brian T. Pentland and Martha S. Feldman. Designing routines: On the folly of designing artifacts, while hoping for patterns of action. *Information and Organization*, 18(4):235–250, 2008.
- [30] Brian T. Pentland, Martha S. Feldman, Markus C. Becker, and Peng Liu. Dynamics of organizational routines: A generative model. *Journal of Management Studies*, 49(8):1484–1508, 2012.
- [31] Brian T. Pentland, Thorvald Hærem, and Derek Hillison. Comparing organizational routines as recurrent patterns of action. *Organization Studies*, 31(7):917–940, 2010.
- [32] Brian T. Pentland, Thorvald Hærem, and Derek Hillison. The (n)ever changing world: Stability and change in organizational routines. *Organization Science*, 22(6):1369–1383, 2011.

- [33] Matteo Priorelli, Ivilin Peev Stoianov, and Giovanni Pezzulo. Embodied decisions as active inference. *PLOS Computational Biology*, 21(6):e1013180, 2025.
- [34] Mikhail I. Rabinovich, Valentin S. Afraimovich, Christian Blick, and Pablo Varona. Information flow dynamics in the brain. *Physics of Life Reviews*, 9(1):51–73, 2012.
- [35] Carolina Rojas-Córdova, Amanda J. Williamson, Julio A. Pertuze, and Gustavo Calvo. Why one strategy does not fit all: A systematic review on exploration–exploitation in different organizational archetypes. *Review of Managerial Science*, 17(7):2251–2295, 2023.
- [36] Christian Stadler, Tazeeb Rajwani, and Florence Karaba. Solutions to the exploration/exploitation dilemma: Networks as a new level of analysis. *International Journal of Management Reviews*, 16(2):172–193, 2014.
- [37] Haridimos Tsoukas and Efi Vladimirov. What is organizational knowledge? *Journal of Management Studies*, 38(7):973–993, 2001.
- [38] Friedrich A. von Hayek. *The Sensory Order*. Routledge and Kegan Paul, London, 1952.